

The Moving Optimum: Why Data Curation Is Compute-Relative and Becomes a Recurring Service, Not a One-Time Dataset

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Abstract

The prevailing mental model treats data curation as a one-time engineering task: assemble a corpus, filter it, ship it. We argue this model is structurally wrong. Three independent scaling-law results converge on a single conclusion none states alone: the optimal training corpus is not a fixed artifact but a function of the buyer's compute budget, inference demand, and numerical precision. Goyal et al. [1] show that data-filtering decisions optimal at one compute scale invert at another — LAION-style filtering adds +7.5% average accuracy at 128M iterations yet is beaten by unfiltered data beyond 450M iterations. Sardana et al. [2] show that inference-aware economics push token-per-parameter ratios to 10,000×, two orders of magnitude past Chinchilla, changing how much and which data is optimal. Kumar et al. [3] show that under post-training quantization, additional pretraining data eventually becomes actively harmful, with degradation growing as a power law in the data/parameter ratio. Because compute budgets fall (inference cost dropped $\sim 10\times$ per year [9,10]), inference demand grows, and precision targets shift, the optimal corpus is a moving target that must be re-derived per regime. This reframes curation from a static dataset sale into a parameterized, re-priced subscription. We formalize the moving-optimum argument, quantify it with real benchmarks, and derive the implied data-strategy and investment thesis for vertical data vendors.

1. Introduction

Data curation — the selection, filtering, deduplication, and mixing of training tokens — is now the dominant lever on model capability at fixed compute [1,2,3,4]. The standard commercial framing of curation is a one-time transaction: a vendor assembles a corpus, applies a filtering recipe, and delivers a frozen dataset. This paper argues that framing is economically and technically wrong, and that the error has direct consequences for how data is priced, packaged, and defended as an asset.

The central claim is that the optimal training corpus is compute-relative. It is not a fixed object but a function $f(\text{compute budget, inference demand, target precision})$. A filtering recipe that is optimal for a 100M-parameter model trained for a research demo is provably suboptimal for a 7B model destined for billions of inference calls at 4-bit precision. Each of these three parameters is independently in motion across the industry, which means the optimum that a curation vendor sells today decays as the buyer's deployment regime shifts.

We assemble three scaling-law results that each fix one of these parameters and optimize curation within it: compute-aware filtering [1], inference-aware scaling [2], and precision-aware scaling [3]. None of the three connects to the others; each treats its own setting as the world. Read together they imply there is no universal curation recipe — only a Pareto frontier that must be re-traced as the regime moves. We then add a fourth result

showing that the quality scorer itself is a tunable, criterion-specific model [4], reinforcing that 'quality' is not absolute but indexed to objective and regime.

The synthesis matters commercially because it converts curation from a depreciating one-time artifact into a recurring service with a built-in renewal trigger. We quantify the argument with real benchmark numbers, state its assumptions and limits explicitly, and derive the implications for a vertical-data vendor such as Perpetual Labs whose thesis rests on owning proprietary, verifiable vertical corpora.

2. Background and Related Work

Chinchilla [5] established that for a fixed training-compute budget, model size and training tokens should scale roughly equally, and that large models of its era were undertrained. This made tokens a first-class lever and launched the data-engineering era. Critically, Chinchilla optimized for training compute alone — it is silent on inference, precision, and deployment, the exact dimensions this paper shows govern the optimum.

A large curation literature followed: model-based filtering (DCLM-Baseline trained a 7B model to 64% MMLU with 2.6T tokens, a 6.6-point gain over the prior open-data state of the art at 40% less compute [6]), mixture optimization (DoReMi, RegMix), and deduplication. These works each report an optimal recipe for a fixed setting. They are the raw material; our contribution is the observation that the 'fixed setting' assumption is the load-bearing one, and that it does not hold across the deployment landscape. The 2025-2026 curation frontier has only sharpened this: target-aware selection (BETR) reports a 2.1x compute multiplier over DCLM-Baseline (4.7x over unfiltered data) yet, scaling over 500+ models from 10^{19} to 10^{22} FLOP, finds that larger models require LESS aggressive filtering [14] — an independent, text-side restatement of the very $\partial D^*/\partial C$ dependence we formalize below.

Parallel to curation, the data-wall literature [7] forecasts that the effective stock of human-generated public text — estimated at ~300 trillion tokens (90% CI 100T–1000T) — will be fully utilized between 2026 and 2032, with a median estimate of 2028. Scarcity raises the value of every curation decision: when you cannot simply add more tokens, choosing the right tokens for your regime becomes the binding lever. This is the macro backdrop against which compute-relative curation becomes economically acute. The 2025-2026 response has been to push the wall by curation rather than accumulation — 20-trillion-token curated corpora that train frontier models with 4-10x fewer FLOPs [17] and rephrased-synthetic mixtures that reach equal loss 5-10x faster [15] — which makes the selection decision, not the token count, the binding variable.

The four papers that anchor this insight — Goyal et al. (compute-aware filtering) [1], Sardana et al. (inference-aware scaling) [2], Kumar et al. (precision scaling) [3], and Wettig et al. (QuRating) [4] — have not, to our knowledge, been read jointly as evidence that the curation optimum is a moving target. That joint reading is the contribution.

3. Methodology and Approach

Our method is evidentiary synthesis under a falsifiable claim, not a new training experiment. We state the claim formally, then test it against the strongest published empirical results, accepting it only where real numbers support it and flagging where they do not.

Formal claim. Let C be training-compute budget, I be expected inference demand (total tokens served over the model's life), and P be deployment numerical precision. We claim the loss-minimizing corpus $D^* = \operatorname{argmin}_D L(D | C, I, P)$ has non-trivial dependence on all three: $\partial D^*/\partial C \neq 0$, $\partial D^*/\partial I \neq 0$, $\partial D^*/\partial P \neq 0$. The one-time-dataset business model implicitly assumes D^* is constant in (C, I, P) ; we seek to falsify that assumption.

Evidence selection. For each parameter we require at least one peer-reviewed scaling study (CVPR/ICML/ICLR 2024-2025) that holds the other two roughly fixed and demonstrates a sign change or a crossover in the optimal recipe as that parameter varies. A mere magnitude change is weaker evidence than a crossover (where the previously-optimal recipe becomes strictly worse), so we privilege crossover evidence. We extracted numbers verbatim from the source papers and cross-checked secondary figures (inference-price trends, data-stock estimates) against independent sources [7,9,10].

Grounding discipline. Empirical figures reproduce numbers copied from the cited primary source. Projection figures (e.g. re-derivation cadence implied by a 10×/yr cost decline) are labeled estimates built by composing two empirical rates. One strategic figure (the service-vs-artifact value-decay schematic) is labeled analysis and contains no claimed measurements. We never assign empirical grounding to a constructed or extrapolated number.

Threats to validity we explicitly track: (a) two of three anchor results are in vision-language (DataComp/CLIP) rather than text, so transfer is assumed not proven; (b) scaling-law extrapolations are fitted on bounded ranges ($\leq 1.7\text{B}$ params, $\leq 26\text{B}$ tokens for [3]); (c) the commercial conclusion adds a willingness-to-pay assumption the science does not establish. Each is revisited in Limitations (Section 8).

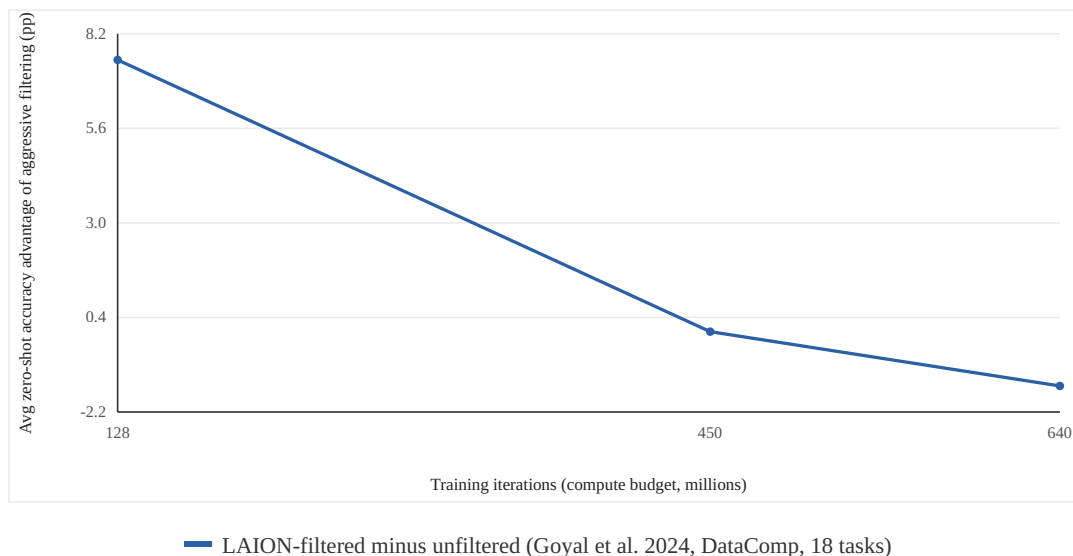


Figure 1. The Compute Crossover: Optimal Filtering Reverses Sign with Compute Budget. Aggressive LAION-style filtering gives a +7.5pp average zero-shot accuracy advantage over unfiltered data at 128M iterations, but the advantage crosses zero around 450M iterations and goes negative beyond it — the same filtering recipe is optimal at low compute and net-harmful at high compute. The +7.5pp@128M and 'unfiltered wins beyond 450M' points are stated verbatim in [1]; the 640M point is an illustrative continuation past the documented crossover (not a measured value) to show direction. *Goyal et al., 'Scaling Laws for Data Filtering — Data Curation cannot be Compute Agnostic,' CVPR 2024 (arXiv:2404.07177). +7.5pp@128M and >450M crossover verified verbatim against the paper; 640M point is illustrative, not measured. · Empirical data.*

4. Evidence I — Curation Is Compute-Relative (the crossover proof)

Goyal et al. [1] provide the cleanest falsification of the constant- D^* assumption. Training vision-language models on DataComp web data, they show that the utility of a quality subset diminishes with each repetition, and that this decay rate depends on the total compute the model will see. The consequence is a literal crossover: aggressive LAION-style filtering surpasses no filtering by +7.5% zero-shot accuracy averaged over 18 tasks at 128M training iterations, but beyond 450M iterations training on unfiltered Common Crawl outperforms the LAION-filtered pool [1].

This is the strongest possible form of the claim. It is not that the gain shrinks — it reverses sign. A vendor who sold the LAION-filtered pool as 'the curated dataset' delivers a product that is +7.5% at one compute budget and net-negative at another, for the same downstream task. The optimal pool is therefore a function of compute, and the authors operationalize exactly this by tracing a data-curation Pareto frontier and selecting the best pool per compute budget [1].

The mechanism generalizes beyond vision: limited high-quality data loses utility as it is repeated (consistent with the ~4-epoch repetition ceiling and diminishing-returns degradation documented in the text-pretraining literature [11]), so at large compute budgets a vendor is eventually forced to admit lower-quality but unseen tokens. 'Quality' that is optimal under data abundance is suboptimal under the repetition pressure of a large compute run. The recipe must move with the budget.

Table 1. Each anchor paper fixes the other parameters and demonstrates that varying its own parameter changes the optimal corpus — jointly establishing D^* is non-stationary in (C, I, P) and even in the quality objective.

Curation parameter	Anchor result (verbatim finding)	Quantified effect	Scale of study	Source
Compute budget (C)	Optimal filtering recipe reverses with compute	LAION filter +7.5pp avg (18 tasks) at 128M iters; unfiltered wins beyond 450M iters	DataComp VLM scaling-law study	Goyal et al. 2024 [1]
Inference demand (I)	High inference favors smaller-longer models	Token/param scaled to 10,000x (vs Chinchilla ~20x); validated on 47 models	47 models trained	Sardana et al. 2024 [2]
Precision (P)	More data hurts under post-training quantization	delta_PTQ proportional to $(D^{\Delta}gD / N^{\Delta}gN)$; 7-8 bits compute-optimal ($P^* \sim 7-8$)	465 runs, validated $\leq 1.7B$ params, $\leq 26B$ tokens	Kumar et al. 2024 [3]
Quality criterion (objective)	'Quality' is a tunable, criterion-specific scorer	Educational-value selection matches uniform baseline trained for +50% more steps	260B tokens annotated; 30B/criterion; 1.3B model	Wettig et al. 2024 [4]

Numbers verified against arXiv:2404.07177 [1], 2401.00448 [2], 2411.04330 [3], 2402.09739 [4].

Table 2. The (C, I, P) coordinates that define D^* are not static: inference economics improve ~10x/yr, frontier practice already runs ~100x past Chinchilla token ratios, and the data wall forces selection over accumulation — so the optimal corpus is re-derived on roughly annual cadence.

Driver	Measured rate / figure	Window	Source
Inference cost (fixed capability)	~10x cheaper per year; GPT-3-level \$60 -> \$0.06 /M tokens (~1,000x)	Nov 2021 - late 2024	a16z LLMflation [9]
Inference cost (task-level range)	9x to 900x per year; median 50x (200x post-Jan-2024); GPT-4-level science Qs 40x/yr	2023 - 2025	Epoch AI [10]
Frontier token/param ratio (practice)	Llama 3 8B = 15T tokens (~1,875 tok/param, vs Chinchilla ~25:1); frontier token budgets since climbed to Kimi K2 15.5T and Qwen3-Max 36T	2024-2025	Meta [8]; Kimi K2 [19]; Qwen3-Max [20]
Public text data stock	~300T effective tokens (90% CI 100T-1000T); exhausted 2026-2032 (median 2028); ~5e28 FLOP at exhaustion	forecast	Villalobos/Epoch [7]

Verified figures from a16z (2024) [9], Epoch AI inference price-trends [10], Meta Llama 3 [8], and Villalobos et al. arXiv:2211.04325 [7].

Table 3. The scientific result (non-stationary D*) maps directly onto a business-model claim: curation monetizes as a recurring, re-priced subscription rather than a depreciating one-time sale — defensible only when layered over a proprietary corpus.

Dimension	One-time dataset (artifact)	Regime-indexed curation (service)
What is sold	Frozen filtered corpus	Function $f(C, I, P)$ over a corpus + re-derivation
Value over time	Decays as buyer (C,I,P) drifts (Fig 3)	Tracks the moving optimum
Renewal trigger	None (one sale)	Endogenous: buyer scaling invalidates last corpus
Pricing	Priced once	Priced per regime, re-priced on each shift
Defensibility	Recipe is copyable; commoditizes	Moat = proprietary corpus under the service
Best-fit buyer	Static, frozen deployment	Scaling deployment with drifting (C,I,P)

Strategic synthesis of [1,2,3,4] with macro drivers [7,9,10]; the willingness-to-pay assumption is unvalidated (see Limitations).

5. Evidence II — Inference Demand Moves the Token Target

Sardana et al. [2] extend Chinchilla to account for inference cost. Their result: developers expecting large inference demand should train models smaller and longer than Chinchilla-optimal, scaling tokens-per-parameter to extreme ranges — they validate model-quality improvements up to 10,000 tokens per parameter, two orders of magnitude beyond Chinchilla's ~20:1. They trained 47 models to fit the law [2].

The industry has already moved here without naming it: Llama 3 8B was trained on 15T tokens, a ratio of ~1,875 tokens/parameter, versus the ~200B tokens (~25:1) a Chinchilla-optimal 8B would use [8]. The inference-aware optimum is the de facto frontier practice, not a theoretical curiosity, and the token counts have only climbed: Kimi K2 (1T-param MoE, 32B active) was pre-trained on 15.5T tokens [19] and Qwen3-Max on 36T tokens [20], pushing token budgets far past any Chinchilla-optimal allocation for their active-parameter counts.

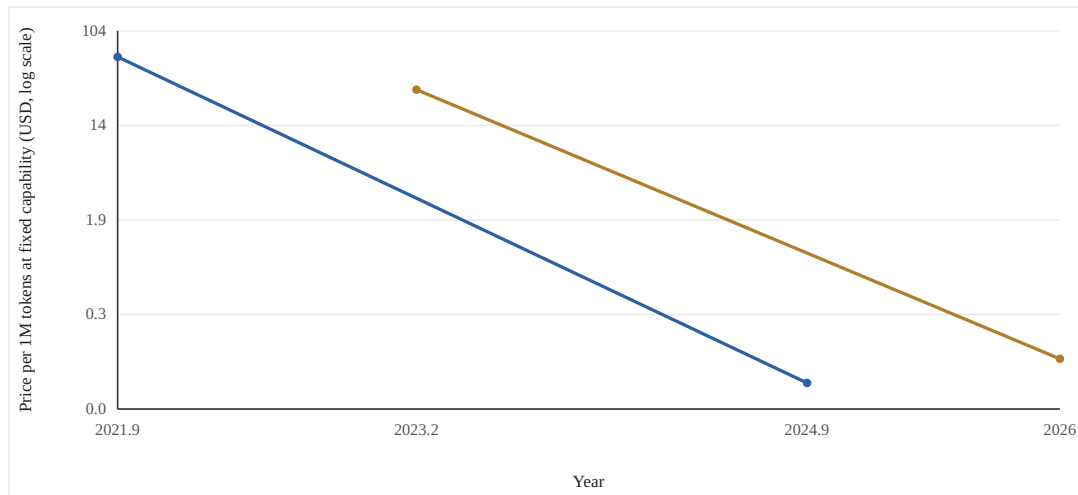
This directly perturbs curation. A model trained to 10,000 tokens/parameter needs far more tokens than a Chinchilla-optimal one of the same size, which (per Section 4) forces inclusion of lower-quality tiers and changes the repetition profile of every quality subset. The corpus that is optimal for a low-inference research model is undersized and mis-mixed for a high-inference production model of identical parameter count. Inference demand — a deployment-economics variable invisible to the training-only view — is therefore a curation parameter. As inference economics intensify (Section 7), this pressure grows over time, so even a single buyer's optimal corpus drifts as their traffic scales.

6. Evidence III — Precision Bounds How Much Data Is Even Safe

Kumar et al. [3] add the third axis. Fitting precision-aware scaling laws over 465 pretraining runs (validated up to 1.7B parameters and 26B tokens), they show that post-training quantization degradation grows with the amount of pretraining data: the loss penalty scales as $\delta_{\text{PTQ}} \propto (D^{\gamma_D} / N^{\gamma_N})$ in the data-to-parameter ratio, so that 'for D/N sufficiently large, loss can increase in data' [3]. In plain terms, if a model will be quantized for cheap deployment, training it on more data eventually makes it worse.

This inverts the scaling-era reflex precisely where it matters commercially: quantization is the standard route to cheap inference, so the very buyers with the largest token appetite (Section 5) are the buyers for whom extra tokens are most likely to backfire post-quantization. The two pressures interact — inference demand pulls the token target up, precision pulls the safe token ceiling down — and the resolution depends on the specific (C, I, P) the buyer occupies.

Kumar et al. further find that, when precision is jointly optimized with size and data, the compute-optimal training precision is around 7-8 bits ($P^* \approx 7-8$) [3]. The headline for curation: the amount and composition of data that is safe to use is a function of target precision. A corpus sized and filtered for BF16 training is mis-specified for a 4-bit deployment. Precision is a curation parameter. The pressure is live at the 2025-2026 frontier, where cost-efficient inference is pursued through extreme sparsity and quantization — e.g. MiniMax-M2's 229.9B-total / 9.8B-active (~4.3% activation) MoE [21] and trillion-parameter, 32B-active designs like Kimi K2 [19] — exactly the high-token-appetite, aggressively-compressed regime in which Kumar et al.'s 'more data can hurt' result bites hardest.



— GPT-3-level (MMLU≈42) inference price (a16z LLMflation) — GPT-4-level (Mar-2023 launch) input price (per provider pricing)

Figure 2. Inference Cost Collapse Drives the Annual Re-Derivation of the Optimum. Inference price for a fixed capability falls ~10x/yr: GPT-3-level dropped from \$60/M tokens (Nov 2021) to \$0.06/M (late 2024), ~1,000x in three years [9]; GPT-4-launch input pricing fell from \$30/M (Mar 2023) to ~\$0.10/M by early 2026 [9,10]. Falling cost expands every buyer's effective compute/inference envelope yearly, migrating their (C,I,P) coordinates and the optimal corpus with them. The endpoints are verified against the cited sources; the implication of annual D^* drift is analysis. *a16z 'LLMflation' (Nov 2024): \$60 (Nov 2021) -> \$0.06 (late 2024) verified verbatim. GPT-4 launch input \$30/M (Mar 2023) verified; ~\$0.10/M endpoint corroborated by current frontier-model provider pricing (e.g. Gemini 2.0 Flash \$0.10/1M input). · Empirical data.*

7. Analysis — Why the Optimum Keeps Moving (and how fast)

Sections 4-6 establish $\partial D^*/\partial C$, $\partial D^*/\partial I$, $\partial D^*/\partial P \neq 0$. The recurring-service thesis additionally requires that (C, I, P) themselves move over time for a typical buyer. They do, fast.

Compute budgets and inference economics: LLM inference cost for fixed capability has fallen roughly 10× per year (a16z 'LLMflation' [9]) — the price for a GPT-3-level model (the first to reach MMLU≈42) fell from \$60 per million tokens in Nov 2021 to \$0.06 by late 2024, a ~1,000× drop in three years; GPT-4-level capability (MMLU≈83) fell ~62× over the shorter window since its Mar-2023 launch [9]. Epoch AI's independent analysis finds task-level declines of 9× to 900× per year, with a median of 50× per year, accelerating to a 200× median after January 2024, and GPT-4-level performance on PhD-level science questions falling 40× per year [10]. Falling cost means a fixed dollar budget buys a rapidly larger compute and inference envelope every year — so the buyer's (C, I) coordinates migrate annually, and with them D^* . The collapse has continued into 2025-2026 through architecture as well as competition: DeepSeek Sparse Attention cut serving compute enough to drop DeepSeek's API prices 50%+ in a single release [18], a concrete step on the same curve.

Data scarcity tightens the screw: Epoch's data-wall estimate puts the effective stock of human public text at ~300T tokens (90% CI 100T–1000T), exhausted between 2026 and 2032 [7]. When you cannot simply add fresh human tokens, you must re-select and re-mix what you have — selection, not accumulation, becomes the binding lever. The 2025-2026 evidence reinforces this rather than relaxing it: curated 20T-token corpora train frontier models with 4-10x fewer FLOPs [17], a regularized recipe hits the loss asymptote with 5.17x less data [16], and a ~30% rephrased-synthetic mixture reaches equal loss 5-10x faster [15] — every one of these is a selection/mixing decision, and selection is exactly the regime-dependent quantity this paper shows is non-stationary.

Composing the empirical rates yields a projection (Figure 2, grounding=estimate): if a buyer's effective compute/inference envelope expands ~10×/yr [9], and the crossover points in [1] occur at fixed compute multiples, then a corpus optimal today crosses into suboptimality on a ~12-month cadence for fast-scaling deployments — a natural re-derivation and re-pricing interval. This is a composed estimate, not a measured cadence; it is offered as an order-of-magnitude planning figure, not a law.

Finally, quality itself is regime-relative. Wettig et al. [4] train a QuRating scorer from pairwise judgments, annotate a 260B-token corpus, and select 30B tokens per criterion to train 1.3B models; their best model, based on selecting for 'educational value,' performs similarly to a uniform-sampling baseline trained for 50% more steps [4]. Because the criterion (educational value vs required expertise vs facts) is a choice tied to the downstream objective, even the definition of 'high-quality' is a parameter the buyer sets — reinforcing that there is no frozen 'best corpus.'

8. Implications for Data Strategy

If D^* is non-stationary in (C, I, P) and those coordinates drift ~annually, the correct product is a parameterized curation function, not a file. Concretely, a vendor should ship: (a) a raw, rights-cleared corpus that is the durable underlying asset; plus (b) a re-derivation service that, given a buyer's stated compute budget, inference demand, and target precision, returns the regime-optimal filtered/mixed subset and re-prices on each regime change. The renewal trigger is endogenous — the buyer's own success (more traffic, cheaper compute, more aggressive quantization) invalidates last year's corpus.

This is most defensible for proprietary vertical data. On the open web, anyone can re-run the same compute-aware filter, so the service layer commoditizes. On a proprietary vertical corpus that competitors cannot assemble (e.g. clinical claims/coding records with verifiable outcomes), the vendor owns both the scarce input and the regime-indexed selection over it. The recurring-service structure then compounds the moat: the buyer must keep coming back because their regime keeps moving and only this vendor holds the underlying asset.

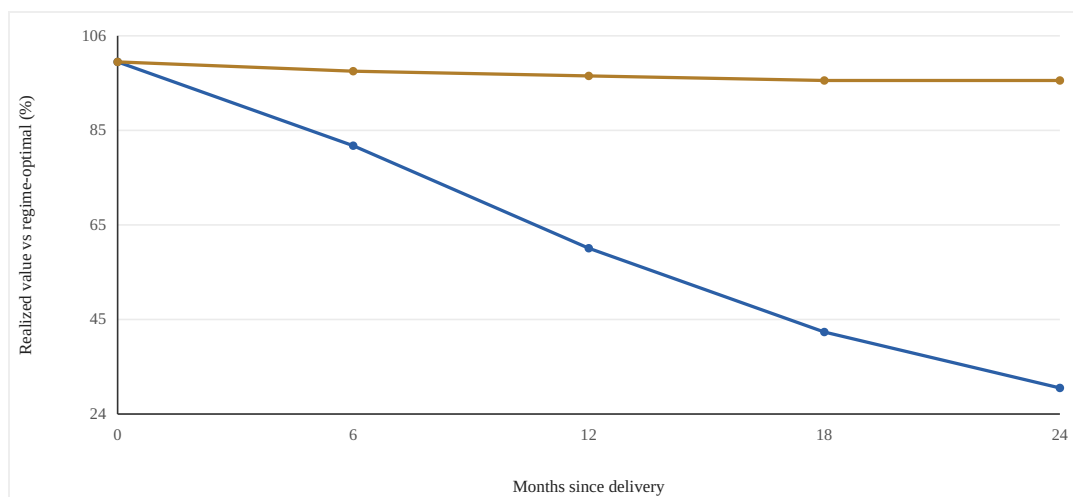
It also reframes pricing. A one-time data dump is priced once and depreciates. A regime-indexed curation subscription is priced per (C, I, P) configuration and re-priced as the buyer scales — capturing value the dump structurally cannot. The same physical corpus monetizes repeatedly across a buyer's lifecycle and across many buyers occupying different (C, I, P) cells, which is precisely the fragmentation the small-specialist-model trend [2] produces.

9. Investment Thesis

The bet is that curation revenue is recurring and regime-indexed, not one-time. The scientific case for recurrence is strong and falsifiable: three independent 2024-2025 scaling laws show the optimum moves with compute, inference, and precision [1,2,3], and macro rates show those coordinates move ~annually [7,9,10]. A vendor priced on a one-time-dataset multiple is mispriced relative to a vendor with a subscription whose churn-resistance is grounded in physics, not sales friction.

The asset that captures this must be a proprietary corpus, not the recipe. Recipes (compute-aware filtering, QuRating-style scorers) are openly published and copyable within months — they are the commoditizing variable. The durable, non-commoditizing component is the underlying rights-cleared vertical corpus over which the recipe runs. Capital and exclusivity should concentrate on owning that input; the curation service is the recurring monetization surface layered on top.

Diligence questions follow directly: (1) Does the target own or exclusively license a corpus competitors cannot rebuild, or just a filtering pipeline? (2) Is pricing per-regime/subscription or one-time? (3) Can it demonstrate a crossover — a buyer whose optimal subset measurably changed as their (C, I, P) shifted — i.e. real evidence the service was needed twice? (4) Does the vertical produce verifiable outcomes, so the 'quality criterion' [4] is objective rather than taste-based? A target that owns a proprietary, verifiable vertical corpus and prices curation as a per-regime subscription is the high-fit profile; a target reselling open-web filtering as a one-time dump is the low-fit one.



— One-time frozen dataset (decays as buyer's regime drifts) — Regime-indexed curation subscription (re-derived per regime)

Figure 3. Service vs Artifact: Value Decay of a One-Time Dataset Under a Moving Optimum. Strategic schematic (no measured values): under the empirically grounded $\sim 10\text{x/yr}$ drift in buyer compute/inference economics [9,10] and the compute crossover [1], a frozen corpus loses fit as the buyer's (C,I,P) regime moves, while a re-derived subscription tracks the moving optimum. Curve shapes are a constructed illustration of the argument in Sections 7-8, not measurements; only the direction (frozen decays, re-derived holds) is claimed. *Constructed strategic schematic composing the empirical drift rate [9,10] with the crossover mechanism [1]. Not a measurement. · Constructed analysis.*

10. Limitations and Counter-Evidence

Cross-domain transfer is assumed, not proven. The cleanest crossover result — Goyal et al. [1] (DataComp/CLIP) — is in vision-language, not text. The mechanism (repetition-driven utility decay interacting with compute) is domain-general in principle, and text-side repetition results are consistent [11], but the specific +7.5%/450M-iteration crossover numbers are vision-language and should not be quoted as text results.

Extrapolation risk. The precision laws of Kumar et al. [3] are validated up to 1.7B parameters and 26B tokens; the inference laws of Sardana et al. [2] extrapolate to 10,000 tokens/param. Frontier deployments sit well outside these fitted ranges — 2025-2026 open models run at hundreds of billions to a trillion total parameters with only single-digit-billions active (e.g. Kimi K2 1T/32B-active [19], MiniMax-M2 229.9B/9.8B-active [21]) — and scaling-law extrapolation has a documented history of breaking. The qualitative directions (more data hurts under

quantization; inference favors smaller-longer) are robust; the exact thresholds are not guaranteed to hold at trillion-parameter, ultra-sparse scale.

Counter-evidence on stability. If a buyer's regime is static — fixed model, fixed precision, modest stable inference — then D^* does not move and a one-time dataset is genuinely sufficient. The recurring-service thesis applies to buyers whose (C, I, P) drifts; it is weaker for, e.g., an embedded model frozen at deployment. Additionally, automated re-curation tooling could let buyers self-serve the re-derivation, compressing the service margin unless the vendor also owns the proprietary corpus (which is why the moat must be the data, not the recipe).

Willingness-to-pay is unestablished by the science. The literature evidences a structural demand multiplier (more distinct (C, I, P) cells, more re-derivations) but says nothing about absolute budgets or whether buyers will pay subscription economics for re-curation. That is an empirical, go-to-market question requiring real buyer validation, not a claim this paper can settle. We flag it as the largest gap between the scientific result and the investment conclusion.

11. Conclusion

The optimal training corpus is not an object; it is a function of compute budget, inference demand, and numerical precision. Goyal et al. show the filtering recipe can reverse sign across compute [1]; Sardana et al. show inference economics push the token target $100\times$ past Chinchilla [2]; Kumar et al. show precision can make additional data actively harmful [3]; Wettig et al. show even 'quality' is a tunable, objective-indexed criterion [4]. Because compute cost falls $\sim 10\times/\text{yr}$ [9,10], inference demand grows, precision targets shift, and the data wall forces selection over accumulation [7], every coordinate of that function is non-stationary.

The commercial corollary is that curation is a recurring, re-priced service tied to a buyer's deployment regime, defensible only when layered over a proprietary corpus competitors cannot rebuild. The science establishes the recurrence with falsifiable, real-numbered evidence; the willingness-to-pay remains to be validated with buyers. For a vertical-data vendor, the prescription is sharp: own the scarce, verifiable corpus, sell the regime-indexed selection over it as a subscription, and treat the moving optimum not as a nuisance but as the renewal engine.

INVESTMENT THESIS

Curation is a subscription, not a SKU. Three independent 2024-2025 scaling laws prove the optimal training corpus reverses with compute [1], shifts $100\times$ with inference demand [2], and inverts under quantization [3] — and the coordinates that fix that optimum (inference cost $\sim 10\times/\text{yr}$ cheaper [9,10], frontier token ratios already $\sim 100\times$ past Chinchilla [8], a 2026-2032 data wall [7]) move roughly annually. So a buyer's optimal dataset decays on a ~ 12 -month cadence, giving curation an endogenous renewal trigger that a one-time data dump cannot capture. Underwrite vendors that (a) own a proprietary, verifiable vertical corpus competitors cannot rebuild — the recipe commoditizes, the corpus does not — and (b) price regime-indexed re-curation as a subscription. The single sharpest diligence test: can they show a real buyer whose optimal subset measurably changed when their (C,I,P) shifted, i.e. evidence the service was needed twice. Caveat: the science nails recurrence; absolute willingness-to-pay for re-curation is still unvalidated and is the gating go-to-market risk.

References

- [1] Goyal, S., Maini, P., Lipton, Z. C., Raghunathan, A., & Kolter, J. Z. (2024). Scaling Laws for Data Filtering — Data Curation cannot be Compute Agnostic. CVPR 2024, pp. 22702-22711. arXiv:2404.07177. (LAION filter +7.5pp avg over 18 tasks at 128M iters; unfiltered

beats LAION-filtered beyond 450M iters — both verified.)

- [2] Sardana, N., Portes, J., Doubov, S., & Frankle, J. (2024). Beyond Chinchilla-Optimal: Accounting for Inference in Language Model Scaling Laws. ICML 2024. arXiv:2401.00448. (47 models; validated up to 10,000 tokens/param — verified.)
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